Complete Practice of Docker with many scenario-based questions part-1

Question 1: Python app works on developer laptop but fails in the lab server

Real-life situation	A startup intern built a Python app on their own machine. It runs locally, but on the shared lab server it fails because the Python version and libraries are different.
Student task	Design a portable container-based solution so the same application runs consistently on all systems.
Expected implementation	Create a Dockerfile, build a custom image, run the container, and verify the app works through terminal or browser.
Syllabus mapping	Introduction to containers, Dockerfile, image layering, container runtime, Docker CLI, image distribution

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```
Terminal
PS C:\Users\sreev> mkdir dockerpractice1/question1


    Directory: C:\Users\sreev\dockerpractice1


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        18-04-2026     09:07                question1


PS C:\Users\sreev> cd dockerpractice1/question1
PS C:\Users\sreev\dockerpractice1\question1> notepad app.py
RAM 2.02 GB  CPU 5.31%   Disk: 47.76 GB used (limit 1006.85 GB)
```

```
Terminal
PS C:\Users\sreev\dockerpractice1\question1> notepad dockerfile
PS C:\Users\sreev\dockerpractice1\question1> docker build -t svish07/flaskapp:1.0 .
[+] Building 6.6s (10/10) FINISHED                                   docker:desktop-linux
 => [internal] load build definition from dockerfile                                  0.1s
 => => transferring dockerfile: 199B                                                  0.0s
 => [internal] load metadata for docker.io/library/python:3.10-slim                   0.0s
 => [internal] load .dockerignore                                                     0.0s
 => => transferring context: 2B                                                       0.0s
 => [1/5] FROM docker.io/library/python:3.10-slim@sha256:26fc00cf5fd6f57ded62e50e27cd3fec9843a3d0f31243c04debb3aba952d9c6  0.1s
 => => resolve docker.io/library/python:3.10-slim@sha256:26fc00cf5fd6f57ded62e50e27cd3fec9843a3d0f31243c04debb3aba952d9c6  0.1s
 => [internal] load build context                                                     0.0s
 => => transferring context: 264B                                                     0.0s
 => CACHED [2/5] WORKDIR /app                                                         0.0s
RAM 2.00 GB  CPU 0.33%   Disk: 47.76 GB used (limit 1006.85 GB)
```

```
Terminal
PS C:\Users\sreev\dockerpractice1\question1> docker login
Authenticating with existing credentials... [Username: svish07]

i Info →To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
PS C:\Users\sreev\dockerpractice1\question1> docker push svish07/flaskapp:1.0
The push refers to repository [docker.io/svish07/flaskapp]
bed04db721b4: Pushed
a6c1adb87640: Pushed
9c930b64e118: Pushed
d145b2bfc575: Pushed
RAM 2.00 GB  CPU 0.42%   Disk: 47.76 GB used (limit 1006.85 GB)
```

```
Terminal
ddf3b231f567: Pushed
1.0: digest: sha256:564b2351b4baa7ee2939e8b9ced0ce0119929b2b10b4af4a9dfc736d1a6a241c size: 856
PS C:\Users\sreev\dockerpractice1\question1> docker run --name flaskapp -p 5000:5000 svish07/flaskapp:1.0
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [18/Apr/2026 03:52:38] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [18/Apr/2026 03:52:38] "GET /favicon.ico HTTP/1.1" 404 -
RAM 2.01 GB  CPU 14.16%  Disk: 47.76 GB used (limit 1006.85 GB)
Terminal  Update available
```

127.0.0.1:5000

Docker runs perfectly!!

```
Terminal
PS C:\Users\sreev\dockerpractice1\question1> type dockerfile
FROM python:3.10-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
EXPOSE 5000
CMD ["python","app.py"]
PS C:\Users\sreev\dockerpractice1\question1> type app.py
from flask import Flask
app=Flask(__name__)

@app.route("/")
RAM 1.99 GB  CPU 0.17%  Disk: 47.76 GB used (limit 1006.85 GB)
Terminal  Update available
```

```
Terminal
COPY . .
EXPOSE 5000
CMD ["python","app.py"]
PS C:\Users\sreev\dockerpractice1\question1> type app.py
from flask import Flask
app=Flask(__name__)

@app.route("/")
def home():
        return "Docker runs perfectly!!"

app.run(host="0.0.0.0",port=5000)
PS C:\Users\sreev\dockerpractice1\question1>
RAM 1.98 GB  CPU 0.50%  Disk: 47.76 GB used (limit 1006.85 GB)
Terminal  Update available
```

Question 2: College wants a standardized coding environment for all students

Real-life situation	In a DevOps lab, every student machine has a different software setup. Faculty wants the same environment for Python, Git, and dependencies on every system.
Student task	Create a reusable Docker image and show how it can be shared using Docker Hub or a registry.
Expected implementation	Build the image, tag it properly, push it to a registry, and pull it on another machine or user account.
Syllabus mapping	Image registries and distribution, Docker registry & Hub, image tagging, Docker architecture

```
Terminal
PS C:\Users\sreev\dockerpractice1> cd question2
PS C:\Users\sreev\dockerpractice1\question2> notepad dockerfile
PS C:\Users\sreev\dockerpractice1\question2> ren dockerfile.txt dockerfile
PS C:\Users\sreev\dockerpractice1\question2> ls

    Directory: C:\Users\sreev\dockerpractice1\question2

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        18-04-2026     09:58            160 dockerfile

RAM 2.45 GB  CPU 4.45%   Disk: 48.13 GB used (limit 1006.85 GB)
```

```
Terminal
PS C:\Users\sreev\dockerpractice1\question2> docker build -t svish07/customenv:1.0 .
[+] Building 46.0s (8/8) FINISHED                                   docker:desktop-linux
 => [internal] load build definition from dockerfile                                0.1s
 => => transferring dockerfile: 200B                                                0.1s
 => [internal] load metadata for docker.io/library/python:3.10-slim                 0.2s
 => [internal] load .dockerignore                                                   0.0s
 => => transferring context: 2B                                                     0.0s
 => [1/4] FROM docker.io/library/python:3.10-slim@sha256:26fc00cf5fd6f57ded62e50e27cd3fec9843a3d0f31243c04debb3aba952d9c6   0.0s
 => => resolve docker.io/library/python:3.10-slim@sha256:26fc00cf5fd6f57ded62e50e27cd3fec9843a3d0f31243c04debb3aba952d9c6   0.0s
 => CACHED [2/4] WORKDIR /workspace                                                 0.0s
 => [3/4] RUN apt-get update && apt-get install -y git vim curl && rm -rf /var/lib/apt/lists/*   32.6s
 => [4/4] WORKDIR /workspace                                                        0.1s
 => exporting to image                                                             12.5s
RAM 2.44 GB  CPU 20.88%   Disk: 48.13 GB used (limit 1006.85 GB)
```

```
Terminal
PS C:\Users\sreev\dockerpractice1\question2> docker login
Authenticating with existing credentials... [Username: svish07]

i Info →To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
PS C:\Users\sreev\dockerpractice1\question2> docker push svish07/customenv:1.0
The push refers to repository [docker.io/svish07/customenv]
4f4fb700ef54: Mounted from svish07/webdemo1-v1
935d60c8480a: Mounted from svish07/flaskapp
ddf3b231f567: Mounted from svish07/flaskapp
bed04db721b4: Mounted from svish07/flaskapp
RAM 2.44 GB  CPU 3.85%   Disk: 48.13 GB used (limit 1006.85 GB)
```

```
1.0: digest: sha256:625effe0d96181039320f85ac1b353ba41e14dcdd47cc69090f45754c1f636dc size: 856
PS C:\Users\sreev\dockerpractice1\question2> type dockerfile
FROM python:3.10-slim
WORKDIR /workspace
RUN apt-get update && apt-get install -y git vim curl && rm -rf /var/lib/apt/lists/*
WORKDIR /workspace
CMD ["bash"]
PS C:\Users\sreev\dockerpractice1\question2>
RAM 2.44 GB  CPU 7.01%   Disk: 48.13 GB used (limit 1006.85 GB)
```

Question 3: Two student teams need isolated environments on the same host

Real-life situation	Two project teams are using the same lab computer. Team A uses Python 3.12 and Team B uses Node.js. They must not disturb each other's setup.

Student task	Design an isolated solution using containers and explain why containers are better than installing everything directly on the host.
Expected implementation	Run separate containers for both teams and show that each has its own runtime and dependencies.
Syllabus mapping	Origin of containers, modern containerization, process isolation, namespaces, container runtime

```
Terminal
PS C:\Users\sreev\dockerpractice1\question3> docker build -t team-a-python .
[+] Building 1.1s (6/6) FINISHED                                   docker:desktop-linux
 => [internal] load build definition from dockerfile                                0.1s
 => => transferring dockerfile: 88B                                                 0.0s
 => [internal] load metadata for docker.io/library/python:3.10-slim                 0.2s
 => [internal] load .dockerignore                                                   0.1s
 => => transferring context: 2B                                                     0.0s
 => [1/2] FROM docker.io/library/python:3.10-slim@sha256:26fc00cf5fd6f57ded62e50e27cd3fec9843a3d0f31243c04debb3aba952d9c6  0.1s
 => => resolve docker.io/library/python:3.10-slim@sha256:26fc00cf5fd6f57ded62e50e27cd3fec9843a3d0f31243c04debb3aba952d9c6  0.1s
 => CACHED [2/2] WORKDIR /app                                                       0.0s
 => exporting to image                                                              0.4s
 => => exporting layers                                                             0.0s
 => => exporting manifest sha256:bc351acce92a824d34c79add3345671c38af6eb5928eb0ec3a7227cd5997a795  0.0s
RAM 2.46 GB  CPU 0.08%   Disk: 48.13 GB used (limit 1006.85 GB)        >_ Terminal   Update available
```

```
Terminal
=> CACHED [2/2] WORKDIR /app                                                        0.0s
=> exporting to image                                                               0.4s
=> => exporting layers                                                              0.0s
=> => exporting manifest sha256:bc351acce92a824d34c79add3345671c38af6eb5928eb0ec3a7227cd5997a795  0.0s
=> => exporting config sha256:9a4f2c2336a49bc9940a7d8c026b5285acf9095ba18cd57acbe54abe2fd4dfff  0.0s
=> => exporting attestation manifest sha256:a6e35a5167cf71bc58e0106eceff884a5a9fc62e9ab36d260958933c6cf624d4  0.1s
=> => exporting manifest list sha256:b5e1bb3787f4144d5e5fc85ac60bb4c55f6b378a33593b35bb5cf8df0f77d9df  0.0s
=> => naming to docker.io/library/team-a-python:latest                              0.0s
=> => unpacking to docker.io/library/team-a-python:latest                           0.0s
S C:\Users\sreev\dockerpractice1\question3> docker run -it --name teamA-python team-a-python
ython 3.10.20 (main, Apr  7 2026, 02:25:42) [GCC 14.2.0] on linux
ype "help", "copyright", "credits" or "license" for more information.
>>
RAM 2.46 GB  CPU 0.42%   Disk: 48.13 GB used (limit 1006.85 GB)        >_ Terminal   Update available
```

```
Terminal
Mode                LastWriteTime         Length Name
----                -------------         ------ ----
-a----        18-04-2026     10:18             40 dockerfile


PS C:\Users\sreev\dockerpractice1\question3\teamB> docker build -t team-b-node .

[+] Building 3.2s (7/7) FINISHED                                   docker:desktop-linux
 => [internal] load build definition from dockerfile                                0.1s
 => => transferring dockerfile: 77B                                                 0.1s
 => [internal] load metadata for docker.io/library/node:18                          2.4s
 => [auth] library/node:pull token for registry-1.docker.io                         0.0s
 => [internal] load .dockerignore                                                   0.1s
RAM 2.47 GB  CPU 0.33%   Disk: 48.13 GB used (limit 1006.85 GB)        >_ Terminal   Update available
```

```
Terminal
 => CACHED [2/2] WORKDIR /app                                                       0.0s
 => exporting to image                                                              0.3s
 => => exporting layers                                                             0.0s
 => => exporting manifest sha256:4696a824a6650524553a69004caaa492656b40a05aa56b700e808ad92288c6f0  0.0s
 => => exporting config sha256:153cd00bbf749f07afa00e26b81385c4a68ec5929db9929f6b136415701d815f  0.0s
 => => exporting attestation manifest sha256:78d330711e047d87f63dc21b8d5e0e55f580ccbc28964f9c59e6eee67524166f  0.1s
 => => exporting manifest list sha256:22f7b3f7b87f481f4d69a554f436b4a48524d556641bb146aede0ce2d96a9332  0.0s
 => => naming to docker.io/library/team-b-node:latest                               0.0s
 => => unpacking to docker.io/library/team-b-node:latest                            0.0s
PS C:\Users\sreev\dockerpractice1\question3\teamB> docker run -it --name teamBnode team-b-node
Welcome to Node.js v18.20.8.
Type ".help" for more information.
>
RAM 2.47 GB  CPU 0.00%   Disk: 48.13 GB used (limit 1006.85 GB)        >_ Terminal   Update available
```

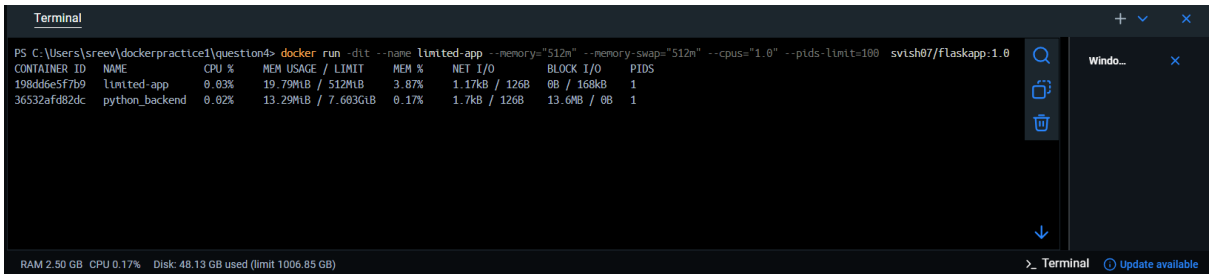
mycontainer	4c397bd72ac0	ubuntu:22.04		0%	2 months ago	
flaskapp	8f796b8b1365	svish07/flaskapp:1.0	5000:5000	0%	1 hour ago	
teamBnode	2978ef3b873c	team-b-node		0%	1 minute ago	
teamApython	3fc5fa2b6029	team-a-python		0%	46 seconds ago	
xai-threat-detection	-	-	-	0%	8 days ago	
frontend_app	07bb5b397618	xai-threat-detection-frontend	3000:80	0%	8 days ago	
dockercomposepractice1-nodem	-	-	-	0%	2 hours ago	
wordpresspract	-	-	-	0%	2 hours ago	

Showing 22 items

```
Terminal
2978ef3b873c1c1ad613c5f2c453ed597d0714c815b30875e9dd0511fd4ef45d
PS C:\Users\sreev\dockerpractice1\question3\teamB> docker run -dit --name teamApython team-b-node
7dbb32d8f1707fe66d6a688634c0782c6b1af4acbc4a6b9c2d549a6056a50829
PS C:\Users\sreev\dockerpractice1\question3\teamB> docker run -dit --name teamApython team-a-python
3fc5fa2b602962131851d9ad07aba0f07a04f8fd188fcbee84eff23925723c52
PS C:\Users\sreev\dockerpractice1\question3\teamB> docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED             STATUS                       PORTS     NAMES
3fc5fa2b6029   team-a-python          "python"                 36 seconds ago      Up 35 seconds                          teamApython
2978ef3b873c   team-b-node            "docker-entrypoint.s…"   About a minute ago  Up About a minute                      teamBnode
835757ee2d48   0236ee02dcbc           "/docker-entrypoint.…"   3 weeks ago         Restarting (1) 6 seconds ago           nginx_gateway
36532afd82dc   dockercompp4-backend   "python app.py"          3 weeks ago         Up 2 hours                             python_backend
ac9a06a43887   0236ee02dcbc           "/docker-entrypoint.…"   3 weeks ago         Restarting (1) 6 seconds ago           gateway
PS C:\Users\sreev\dockerpractice1\question3\teamB>
RAM 2.49 GB  CPU 0.08%   Disk: 48.13 GB used (limit 1006.85 GB)        >_ Terminal   Update available
```

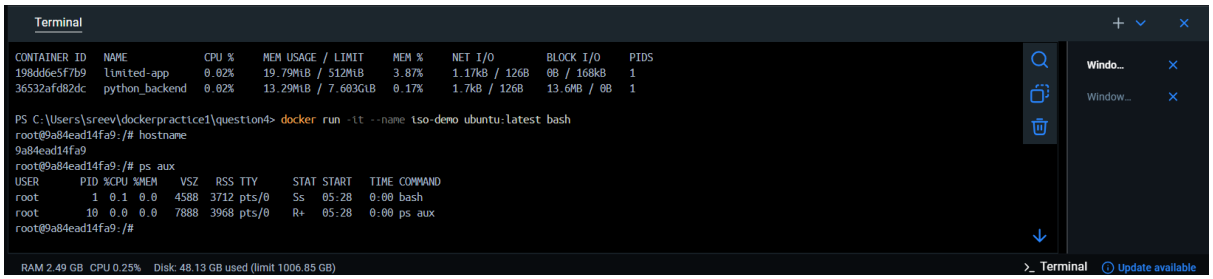
Question 4: A container consumes too much RAM in a shared lab machine

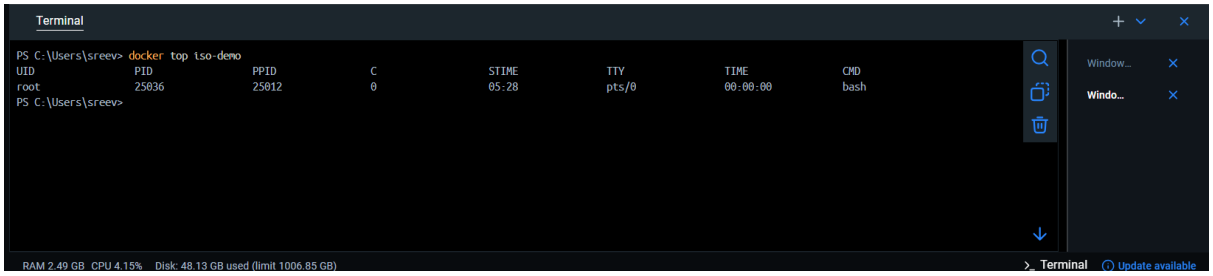
Real-life situation	One student's analytics container starts using excessive memory and slows down all other users on the same system.
Student task	Implement resource control so the container cannot exceed a fixed memory and CPU limit.
Expected implementation	Run the container with memory and CPU limits, monitor it using docker stats, and explain the role of cgroups.
Syllabus mapping	Control groups (cgroups), Docker CLI, container runtime, resource limits



Question 5: Faculty wants students to understand process isolation in real systems

Real-life situation	A faculty member wants proof that containers are isolated processes and not full virtual machines.
Student task	Create a lab demonstration that shows container hostname, PID behavior, and process isolation.
Expected implementation	Run a container interactively, inspect hostname and processes inside it, and compare with host-side output using docker top or inspect.
Syllabus mapping	Process isolation, namespaces, Docker daemon, container runtime





Question 6: A web application loses uploaded files after container deletion

Real-life situation	A student deployed a simple website in a container. After removing and recreating the container, all uploaded content disappeared.
Student task	Design a Docker-based persistence solution so the data survives container recreation.
Expected implementation	Use a Docker volume, mount it to the required path, create sample files, delete the container, and prove the data still exists.
Syllabus mapping	Docker object types - volume, Docker filesystem, writable layer vs persistent storage

Question 7: Frontend container cannot talk to backend container

Real-life situation	A mini project has two containers: one frontend and one backend API. Both are

	running, but the frontend cannot communicate with the backend.
Student task	Design a networking solution so both services communicate correctly and predictably.
Expected implementation	Create a user-defined bridge network, attach both containers, verify name-based communication, and explain internal networking.
Syllabus mapping	Docker network object, Docker architecture, container communication, runtime networking

```
Terminal

PS C:\Users\sreev\dockerpractice1\question4> docker network create mynetw
bba1b1cb155147cfbf53b71f5dc9299c1648eb614d7deda2ce9497121667f36f
PS C:\Users\sreev\dockerpractice1\question4> docker run -dit --name backend --network mynetw -p 5000:5000 svish07/flaskapp:1.0
docker: Error response from daemon: Conflict. The container name "/backend" is already in use by container "c63d31b2bb1a3cfc6ae25235ae3a549301f61fd1a25e46fc0b6c79dcfa4d1
421". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
PS C:\Users\sreev\dockerpractice1\question4> docker run -dit --name backendp --network mynetw -p 5000:5000 svish07/flaskapp:1.0
4b5dca54dd9c7c8a45b954faa0b340ab6a72e03abedf7916cf99a20dca982796
PS C:\Users\sreev\dockerpractice1\question4> docker run -dit --name frontend --network mynetw ubuntu bash
da82383b32d927b3cdc673c61fb10e1c39481f27826585a647ddf114f7b2f743
PS C:\Users\sreev\dockerpractice1\question4> docker exec -it frontend bash
root@da82383b32d9:/# apt update && apt install -y curl

RAM 2.63 GB  CPU 4.84%   Disk: 48.21 GB used (limit 1006.85 GB)
```

```
Terminal

Setting up libssh-4:amd64 (0.10.6-2ubuntu0.4) ...
Setting up libcurl4t64:amd64 (8.5.0-2ubuntu10.8) ...
Setting up curl (8.5.0-2ubuntu10.8) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for ca-certificates (20240203) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
root@da82383b32d9:/# curl http://backend:5000
curl: (6) Could not resolve host: backend
root@da82383b32d9:/# curl http://backendp:5000
Docker runs perfectly!!root@da82383b32d9:/#

RAM 2.64 GB  CPU 5.78%   Disk: 48.21 GB used (limit 1006.85 GB)
```

Question 8: Students must explain why Docker images rebuild slowly

Real-life situation	A student modifies one line in app.py, but the full image rebuild takes too long every time in CI.
Student task	Redesign the Dockerfile so rebuilds are faster and explain which layers can be cached.
Expected implementation	Show a layered Dockerfile, place dependency installation before source copy where appropriate, rebuild, and explain the caching effect.
Syllabus mapping	Container images & layers, Docker layering, Docker filesystem, image build optimization

```
Terminal

-a----        18-04-2026     09:20            160 dockerfile
-a----        18-04-2026     09:18              5 requirements.txt


PS C:\Users\sreev\dockerpractice1\question1> type dockerfile
FROM python:3.10-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
EXPOSE 5000
CMD ["python","app.py"]
PS C:\Users\sreev\dockerpractice1\question1>

RAM 0.95 GB  CPU 0.08%   Disk: 48.21 GB used (limit 1006.85 GB)
```

Question 9: Department wants a ready-to-use demo image for a lab exam

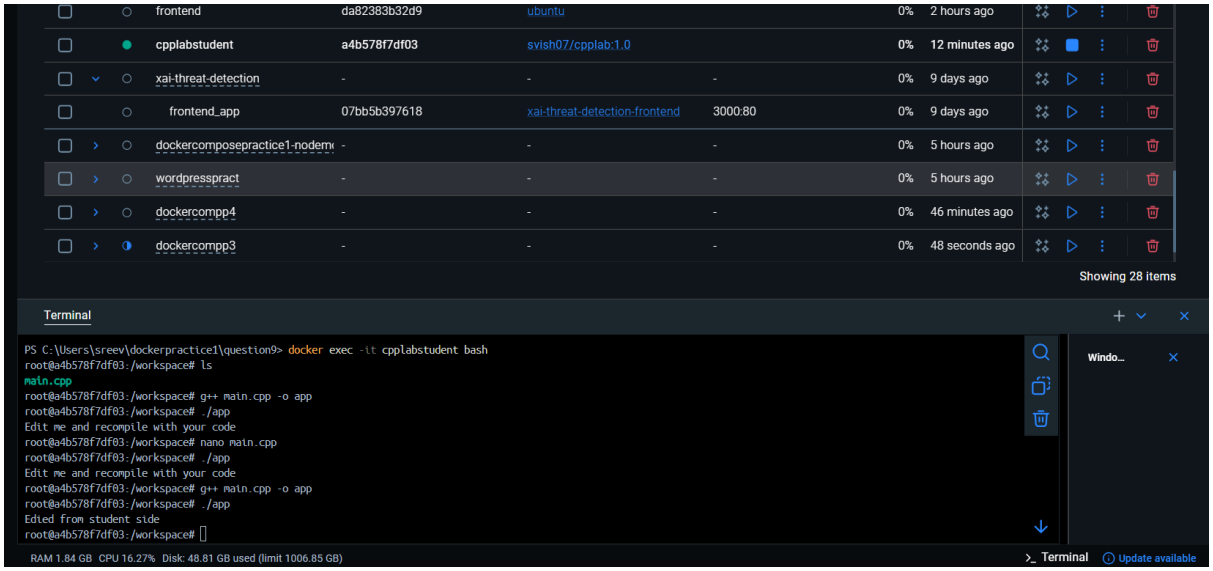
Real-life situation	The department wants a prebuilt image containing tools and code so students can instantly run a practical demo without manual installation.
Student task	Prepare a distributable image and demonstrate how the image is pulled and executed on a clean system.
Expected implementation	Build the image, tag it, optionally push to Docker Hub, pull it again, and run a working container from it.
Syllabus mapping	Image registries and distribution, Docker Hub, Docker daemon, image/container relationship

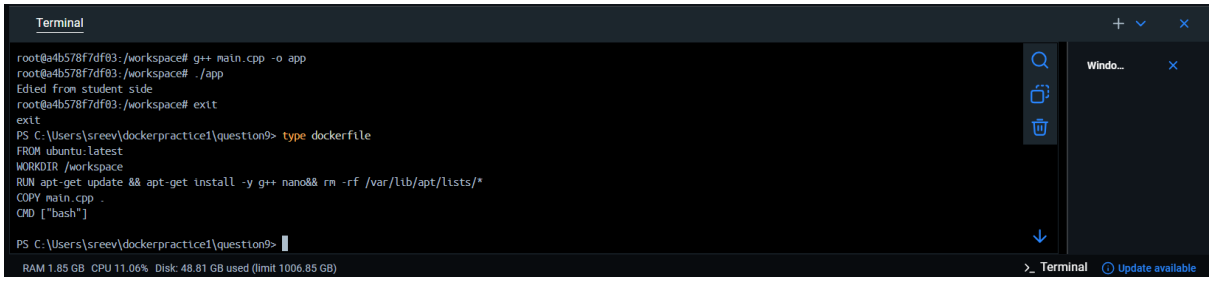
```
Terminal

    Directory: C:\Users\sreev\dockerpractice1\question9

Mode                LastWriteTime         Length Name
----                -------------         ------ ----
-a----        18-04-2026     13:27            154 dockerfile
-a----        18-04-2026     13:24            123 main.cpp


PS C:\Users\sreev\dockerpractice1\question9> docker build -t svish07/cpplab:1.0 .
[+] Building 103.8s (9/9) FINISHED                                   docker:desktop-linux
 => [internal] load build definition from dockerfile                                  0.1s
 => => transferring dockerfile: 193B                                                  0.1s
RAM 1.83 GB  CPU 8.93%  Disk: 48.81 GB used (limit 1006.85 GB)   >_ Terminal  Update available
```


```
Terminal

    Directory: C:\Users\sreev\dockerpractice1\question9

Mode                LastWriteTime         Length Name
----                -------------         ------ ----
-a----        18-04-2026     13:27            154 dockerfile
-a----        18-04-2026     13:24            123 main.cpp


PS C:\Users\sreev\dockerpractice1\question9> docker build -t svish07/cpplab:1.0 .
[+] Building 103.8s (9/9) FINISHED                                   docker:desktop-linux
 => [internal] load build definition from dockerfile                                  0.1s
 => => transferring dockerfile: 193B                                                  0.1s
RAM 1.83 GB  CPU 8.93%  Disk: 48.81 GB used (limit 1006.85 GB)   >_ Terminal  Update available
```


```
Terminal

83270f22e9f3: Pushed
46294977 5cd0: Pushed
1.0: digest: sha256:0f0410d491a045b69295fb33350b814c8de567e0884ee20f4f7ea72bb8c3f9cd size: 856
PS C:\Users\sreev\dockerpractice1\question9> docker rmi svish07/cpplab:1.0
Untagged: svish07/cpplab:1.0
Deleted: sha256:0f0410d491a045b69295fb33350b814c8de567e0884ee20f4f7ea72bb8c3f9cd
PS C:\Users\sreev\dockerpractice1\question9> docker pull svish07/cpplab:1.0
1.0: Pulling from svish07/cpplab
04cd169574d4: Pull complete
f0b7b8c3bc11: Pull complete
83270f22e9f3: Pull complete
46294977 5cd0: Download complete
Digest: sha256:0f0410d491a045b69295fb33350b814c8de567e0884ee20f4f7ea72bb8c3f9cd
RAM 1.83 GB  CPU 13.66%  Disk: 48.81 GB used (limit 1006.85 GB)   >_ Terminal  Update available
```

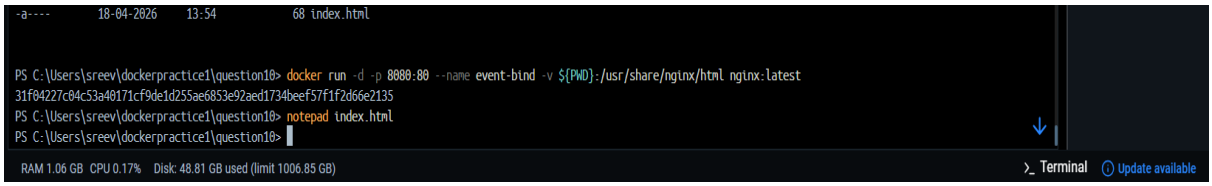
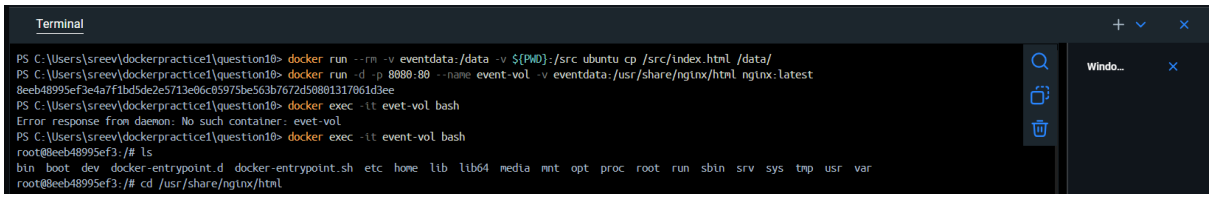
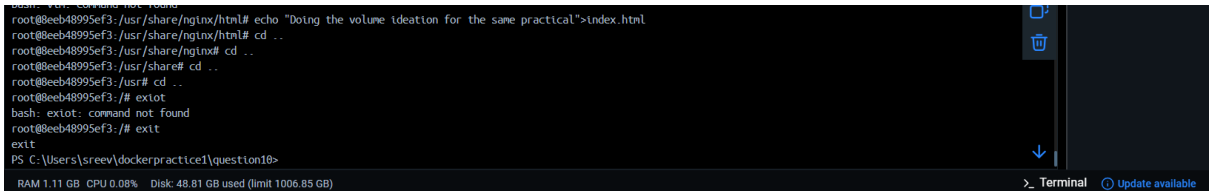
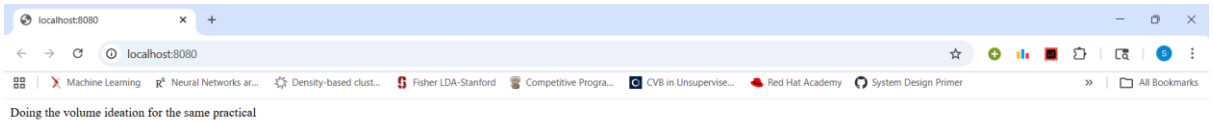
```
Terminal

docker.io/svish07/cpplab:1.0
PS C:\Users\sreev\dockerpractice1\question9> docker run -dit --name cpplabstudent svish07/cpplab:1.0
a4b578f7df03e2df7494f632d452dda4ac230840790b93e4cc16d729cf8e3b3e
PS C:\Users\sreev\dockerpractice1\question9> docker exec -it cpplabstudent bash
root@a4b578f7df03:/workspace# ls
main.cpp
root@a4b578f7df03:/workspace# g++ main.cpp -o app
root@a4b578f7df03:/workspace# ./app
Edit me and recompile with your code
root@a4b578f7df03:/workspace# nano main.cpp
root@a4b578f7df03:/workspace# ./app
Edit me and recompile with your code
root@a4b578f7df03:/workspace# g++ main.cpp -o app
RAM 1.83 GB  CPU 20.41%  Disk: 48.81 GB used (limit 1006.85 GB)   >_ Terminal  Update available
```





Question 10: IT club wants to deploy a static site quickly for an event

Real-life situation	An IT club needs a temporary event website. They want fast deployment, simple rollback, and no manual web server installation on the host machine.
Student task	Design and implement a containerized static web deployment using Docker.
Expected implementation	Create an HTML page, build an Nginx-based image or use bind mount/volume appropriately, publish the port, and test in browser.
Syllabus mapping	Introduction to Docker, containers in DevOps, images, containers, ports, layering, runtime deployment



Though not preferred here as we need fast deployment, but still have done the practical using docker volumes too: